

When do Oil Shocks Move Corn? Evidence for Ethanol-Mediated Transmission in Commodity Markets

Executive Summary

This report examines the transmission of crude oil price shocks to U.S. corn futures, with a particular focus on the role of ethanol-linked energy dynamics.

The analysis conducted found that the evidence for an unconditional relationship between oil and corn returns is weak, with limited predictive power for directional movements. However, oil shocks are associated with increased volatility in corn prices.

Conditional analysis suggests that the oil-corn relationship strengthens during periods of high oil market volatility, alongside periods wherein oil and gasoline markets are more tightly linked. This suggests the existence of a modest transmission channel through energy market conditions.

Broadly, whilst energy markets variably influence corn futures under specific conditions, the relationship is not sufficiently strong or stable to support standalone trading strategies. The findings highlight the importance of regime-based analysis in evaluating cross-commodity relationships, particularly when translating market signals into investment decisions.

Research Questions & Hypothesis

The fundamental research question for the analysis conducted was as follows:

“To what extent do ethanol economics mediate the transmission of oil price shocks into corn futures returns, and under what conditions does this relationship strengthen or break down?”

Furthermore, this report considers the investment-decision oriented question of:

“Can ethanol-related conditions help identify when oil shocks are more likely to translate into actionable insights for corn futures positioning?”

The analysis tests four hypotheses:

Hypothesis 1: Baseline Instability

The relationship between oil and corn returns is unstable over time and not consistently strong.

Hypothesis 2: Shock Transmission

Large oil price shocks exhibit stronger transmission into corn futures returns than smaller price movements, but only under certain market conditions.

Hypothesis 3: Ethanol Conditioning

The oil-corn relationship strengthens when ethanol-related economic conditions are favourable, proxied by a strong gasoline-oil linkage.

Hypothesis 4: Regime Breakdown

The oil-corn relationship weakens during periods dominated by non-energy factors (e.g. agricultural supply shocks).

Data & Methodology

Daily futures data for crude, oil, and gasoline were analyzed from 2000-2026. The data was sourced from Yahoo! Finance (tickers: CL=F for crude oil, ZC=F for corn, and RB=F for gasoline). Any duplicate and irregular values were dropped from each dataset. The three datasets were then aligned by date and merged.

Log returns were computed, and relationships were examined using:

- Correlation analysis.
- Rolling correlations.
- Regime segmentation (volatility and energy linkage).
- Conditional comparisons (shock vs non-shock periods).

Limitations

This analysis does not explicitly incorporate agricultural supply factors (e.g. weather, crop yields), which may dominate corn price dynamics in certain periods.

Additionally, the ethanol proxy used (oil-gasoline correlation) is indirect and may not fully capture biofuel-specific dynamics.

Key Findings

(1) Weak Baseline Relationship

The unconditional correlation between oil and corn returns is low (~ 0.19), and as shown in Figure 1, the relationship appears diffuse and dominated by noise in scatter analysis, with no clear linear structure.

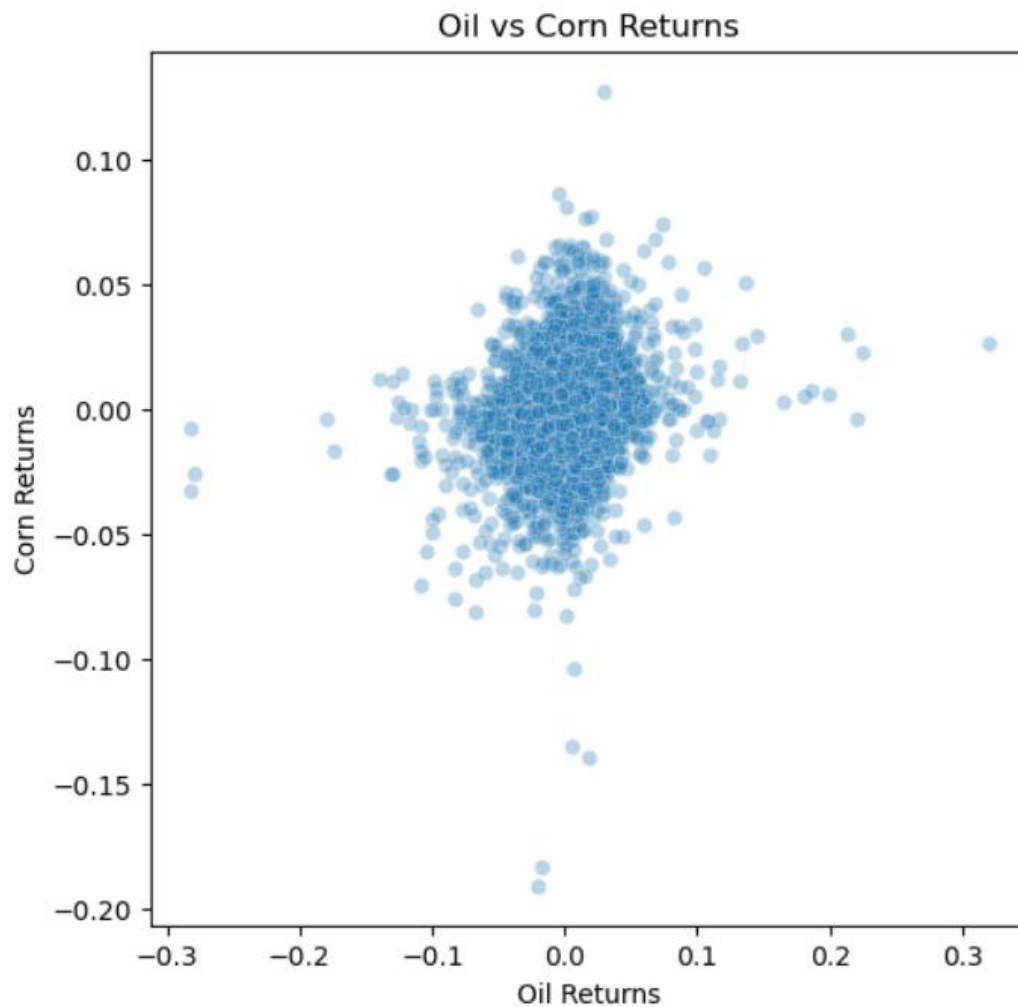


Figure 1: Scatter plot of daily oil vs corn returns showing weak and diffuse relationship.

(2) Oil Shocks Increase Volatility, Not Direction

Oil shock periods do not produce consistent directional moves in corn futures but are associated with higher volatility and larger absolute price movements.

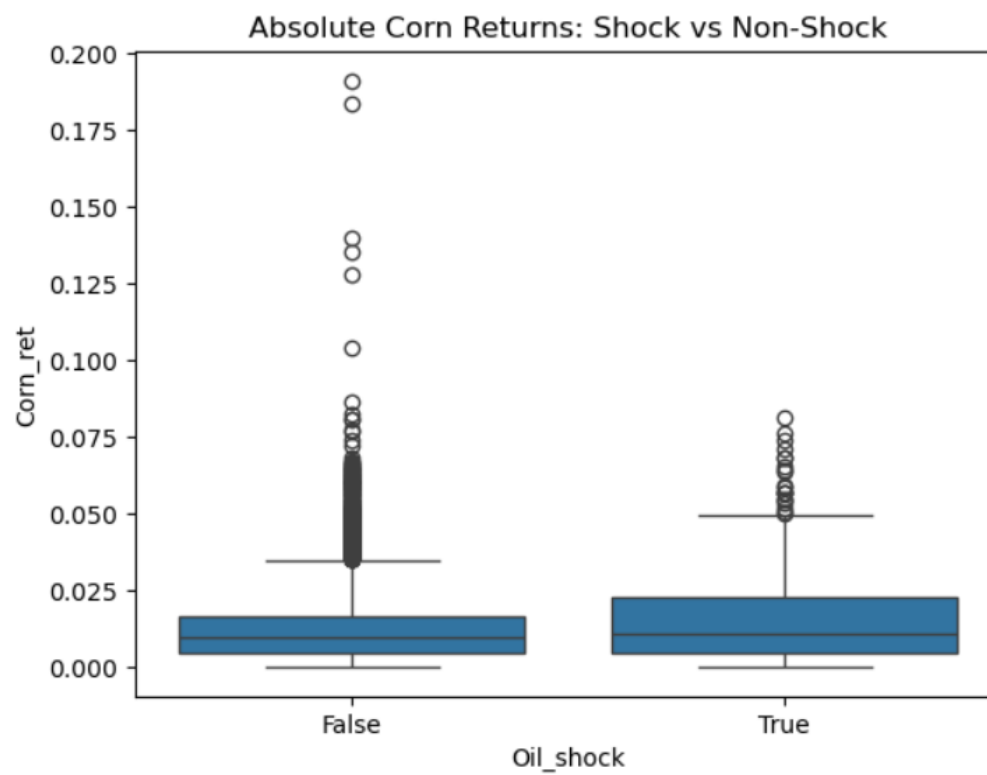
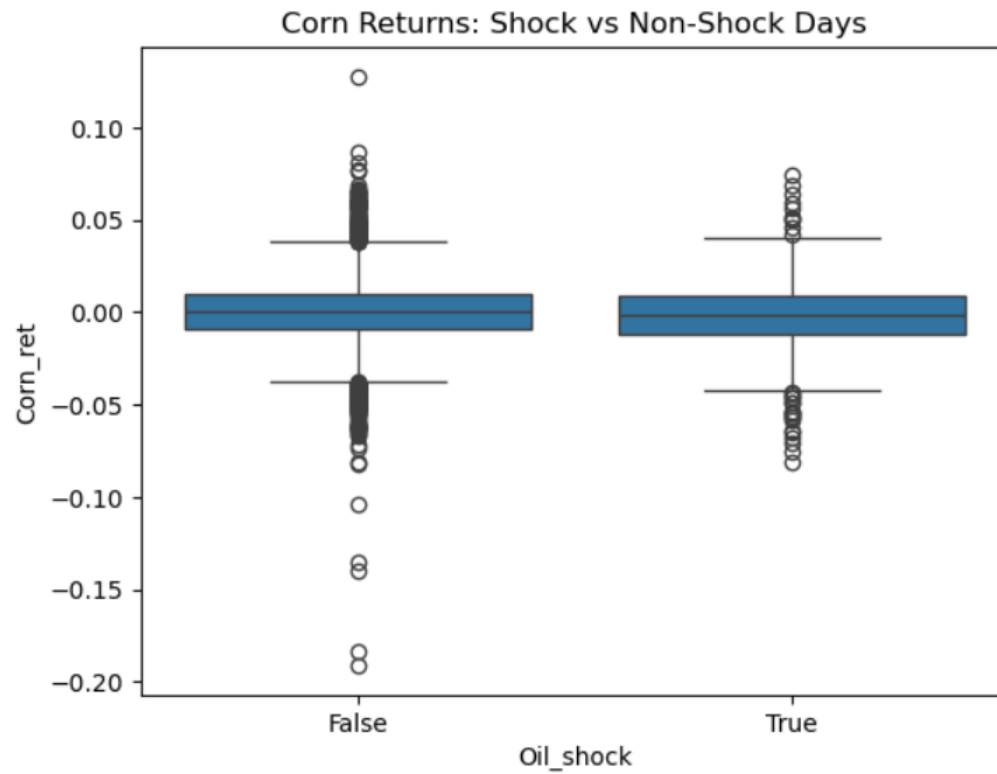


Figure 2: Corn Return distributions show limited directional change but increased dispersion during oil shock periods.

(3) Relationship Strengthens in High-Volatility Regimes

Whilst the baseline relationship is fundamentally weak, conditioning on market regimes reveals a different picture. The oil-corn correlation increases during periods of elevated oil market volatility, suggesting stronger cross-commodity transmission under stress conditions.

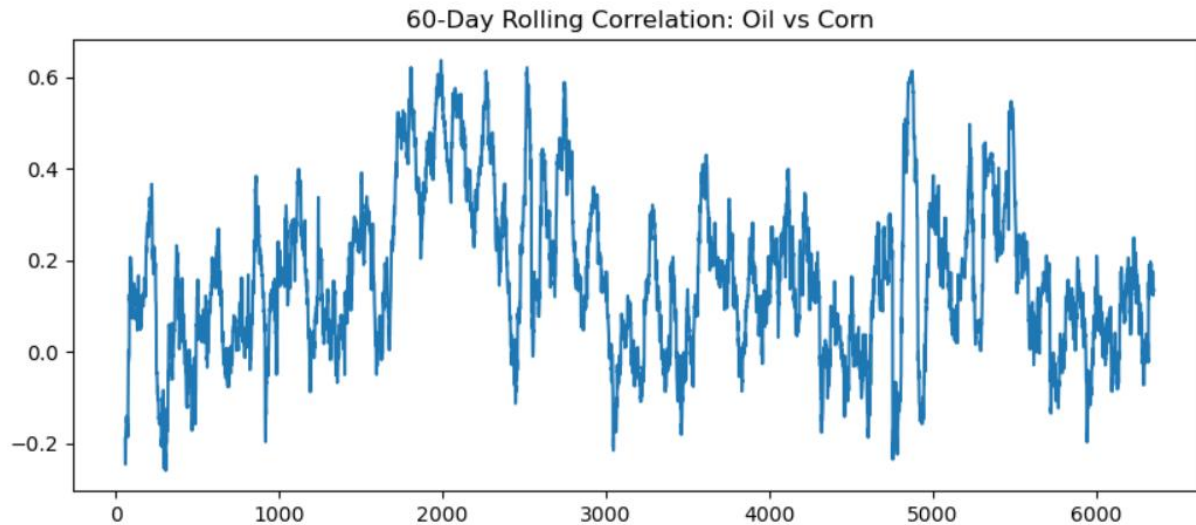


Figure 3: Rolling oil-corn correlation varies significantly over time, indicating regime-dependent relationships.

(4) Evidence of Ethanol-Mediated Linkage

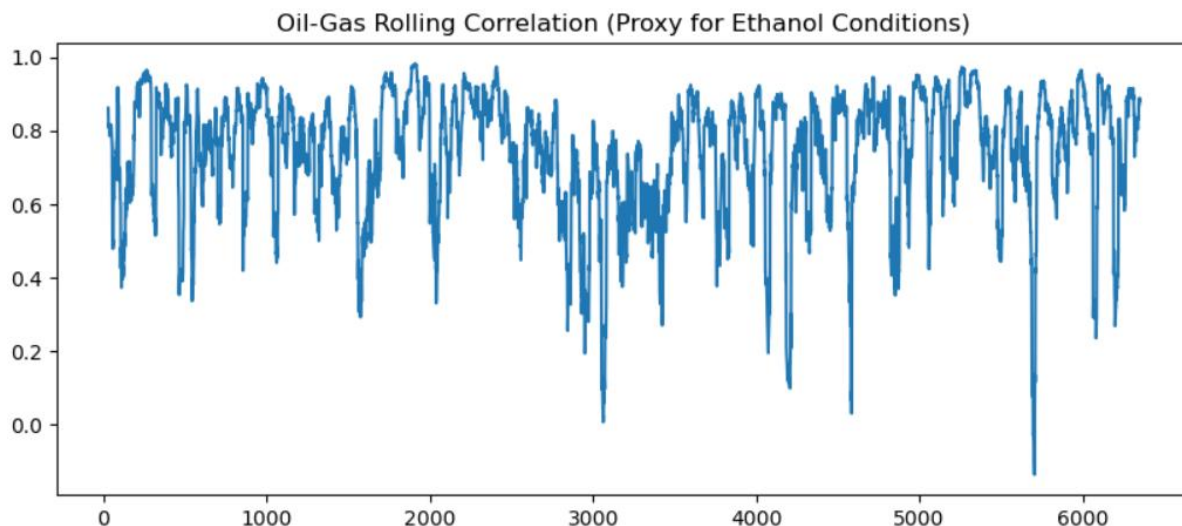


Figure 4: Oil-gasoline correlation used as a proxy for energy market linkage and ethanol conditions.

When oil and gasoline markets are more tightly linked, oil-corn correlation increases from 0.14 to 0.23, alongside modest increases in corn return magnitude and volatility. This suggests a conditional transmission mechanism consistent with ethanol-related dynamics, though the effect is moderate. As Figure 5 demonstrates, this represents a ~65% relative increase in correlation across regimes.

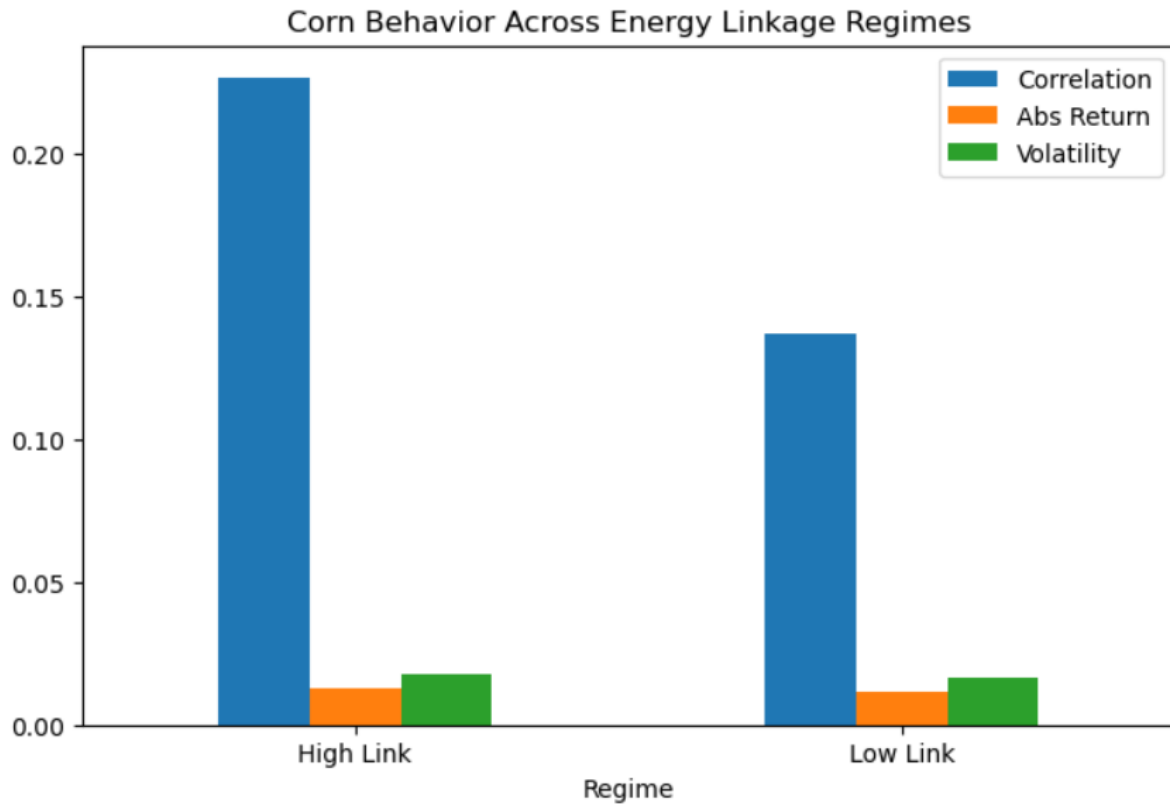


Figure 5: Corn return behaviour across high and low energy linkage regimes, showing higher correlation, magnitude, and volatility when oil-gas relationship is strong.

When each of these findings are considered altogether, they indicate that the oil-corn relationship may not be structurally strong but becomes economically relevant under specific energy market conditions.

Investment Interpretation

The results suggest that oil shocks alone are not reliable signals for positioning in corn futures. However, energy market conditions provide useful context: periods of strong oil-gasoline linkage and elevated volatility are associated with greater sensitivity of corn prices to energy markets.

This necessarily posits that cross-commodity signals should be used conditionally and in conjunction with agricultural fundamentals. Meaning, oil shocks should not be treated as standalone trading signals. Instead, their informational value increases when evaluated within the context of broader energy market conditions, particularly during periods of strong oil-gasoline linkage and elevated volatility.

Conclusion

Whilst ethanol-related dynamics provide a conceptually plausible transmission mechanism between energy and agricultural markets, their pragmatic utility lies in contextual signal refinement, rather than direct trade generation.

Ultimately, oil-corn relationships are conditional, moderate, and unstable over time. Whilst energy markets influence agricultural commodities through mechanisms such as ethanol demand, this transmission is not sufficiently strong to support simple or standalone trading strategies.